Exercise: MCP Anatomy

Background

The mCase MCP server (`@redmane/mcase-mcp-server`) gives Cursor the ability to query live mCase configuration data during a conversation. Understanding what tools it exposes and what data it returns is essential for building the capstone app.

You're going to use the `ai-agent-documentation` skill to understand the codebase rapidly — then use that knowledge to make design decisions.

Step 1 — Generate a Reference Doc (15 min)

The MCaseMCPTools codebase is open in Cursor. Invoke the `ai-agent-documentation` skill:

```
@ai-agent-documentation

Generate a minimal AI reference for this MCP server. Focus on:
1. What tools it exposes and what each one returns (include the data shapes, not
just tool names)
2. The DatalistDefinition and FieldDefinition structures — what fields do they
have and what do they represent?
3. What the templateProcessor filters out and why
4. What's missing from the current tools that a config browser app would want
```

While it generates, open `src/types.ts` to see the raw data shapes.

Step 2 — Understand the Data Model (5 min)

After the reference doc is generated, answer these questions:

Tools:

- Which tool returns a summary list of all datalists?
- Which tool returns the full definition of one datalist (with fields)?
- Which tool returns relationships between datalists?

Data shapes:

- What properties does a `FieldDefinition` have?
- What does `addType` represent? What values might it have?
- What is the `source` property on a FieldDefinition? What does it point to?
- What is the `values` array on a FieldDefinition?

Filtering:

- What kinds of fields does the `templateProcessor` strip out before returning data?
- Why might this be useful for a config browser?

Step 3 — Product Discovery (5 min)

Based on what the tools currently return, answer:

What can your capstone app do well with the current tools?
(List 2-3 things)

What is missing from the current tools that would make the app better?
(List 1-2 gaps)

If you were going to add one tool to the MCP server, what would it be?
Write a one-sentence description: `tool_name — [what it accepts, what it returns]`

Reference: MCP Tool Summary

Tool	Returns
`schema`	`DatalistSummary[]` — Id, Label, SystemName, FieldCount
`all_datalists`	Same as `schema`
`datalist_by_name`	`DatalistDefinition` — full detail including Fields[], Children[]
`datalist_by_id`	Same as `datalist_by_name` using numeric Id
`all_relationships`	`Relationship[]` — TargetDatalist, TargetField, SourceDatalist, SourceField, RelationshipType
`greet`	Smoke test only — ignore

Reference: FieldDefinition Shape

```
{
  fieldId: number
  label: string          // Human-readable name
  systemName: string     // API/code name
  addType: string        // Field type: "Text", "Date", "Picklist", "Dynamic",
etc.
  required: boolean
  hidden: boolean
  isMirror: boolean
  values: string[]       // Picklist options (empty for non-picklist fields)
  source: number         // For Dynamic fields: ID of the linked datalist
  description: string    // Help text (HTML stripped)
}
```